volts. In this manner, the feedback of the circuit forces the capacitor voltage to track the voltage of the input signal. If the input signal changes very rapidly, the voltage on the capacitor changes at a constant rate until a match is obtained. This constant rate of change is called the **slew rate**, just as in other electronic devices such as op amps.

Now, consider the characteristics of the delta modulated output signal. If the analog input is *increasing* in value, the output signal will consist of more ones than zeros. Likewise, if the analog input is *decreasing* in value, the output will consist of more zeros than ones. If the analog input is constant, the digital output will alternate between zero and one with an equal number of each. Put in more general terms, the relative number of ones versus zeros is directly proportional to the *slope* (derivative) of the analog input.

This circuit is a cheap method of transforming an analog signal into a serial stream of ones and zeros for transmission or digital storage. An especially attractive feature is that all the bits have the same meaning, unlike the conventional serial format: *start bit, LSB,⋯ ,MSB, stop bit*. The circuit at the receiver is identical to the feedback portion of the transmitting circuit. Just as the voltage on the capacitor in the transmitting circuit follows the analog input, so does the voltage on the capacitor in the receiving circuit. That is, the capacitor voltage shown in (a) also represents how the reconstructed signal would appear.

A critical limitation of this circuit is the unavoidable tradeoff between (1) maximum slew rate, (2) quantization size, and (3) data rate. In particular, if the maximum slew rate and quantization size are adjusted to acceptable values for voice communication, the data rate ends up in the MHz range. This is too high to be of commercial value. For instance, conventional sampling of a voice signal requires only about 64,000 bits per second.

A solution to this problem is shown in Fig. 3-18, the Continuously Variable Slope Delta (**CVSD**) modulator, a technique implemented in the Motorola MC3518 family. In this approach, the clock rate and the quantization size are set to something acceptable, say 30 kHz, and 2000 levels. This results in a terrible slew rate, which you correct with additional circuitry. In operation, a shift resister continually looks at the last four bits that the system has produced. If the circuit is in a slew rate limited condition, the last four bits will be all ones (positive slope) or all zeros (negative slope). A logic circuit detects this situation and produces an analog signal that increase the level of charge produced by the charge injectors. This boosts the slew rate by increasing the size of the voltage steps being applied to the capacitor.

An analog filter is usually placed between the logic circuitry and the charge injectors. This allows the step size to depend on how long the circuit has been in a slew limited condition. As long as the circuit is slew limited, the step size keeps getting larger and larger. This is often called a *syllabic filter*, since its characteristics depend on the average length of the syllables making up speech. With proper optimization (from the chip manufacturer's